PREP 1, LESSONS 1–2

Particles and gas pressure

Solids, liquids and gases; changes of state; what causes gas pressure

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons, 1 mark each

1 State the frequency and potential difference of the UK mains electricity supply. DN 132 [1]

50 Hz and about 230 V

2 State the function of the earth wire in a three-core cable. DN 136 [1]

It is a safety wire that stops the appliance becoming live.

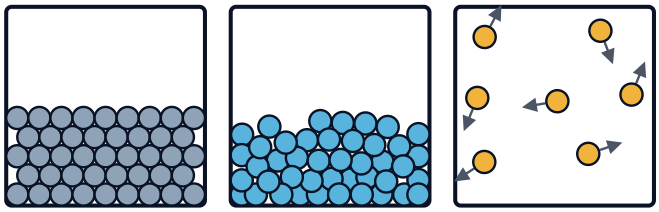
3 State the two measurements needed to find the resistance of a component. DN 108 [1]

The potential difference across it and the current through it

Key ideas LESSONS 1–2

Show your working

4 The diagrams show the particles in a solid, a liquid and a gas.



a The diagram that shows a liquid is A (B) C [1]

b Describe the arrangement and the motion of the particles in a gas. [2]

Far apart, in a random arrangement ✓

moving quickly in all (random) directions ✓

5 A 30 g ice cube melts in a sealed bag. The mass of the water is less than 30 g (30 g) more than 30 g [1]

6 A sealed can of gas is heated. Explain, in terms of particles, why the pressure of the gas increases. [3]

The particles move faster (gain kinetic energy) ✓

They hit the walls more often ✓

and with a bigger force ✓